

MRM empirical callables (OTIS / TPS / SIU)

Overview

This vignette documents the `mrm_otis_*`, `mrm_tps_*`, and `mrm_siu_*` empirical callables. Each function is a one-line entry point to a verified analysis used in the *MRM empirical paper* (Ruhela 2026, in preparation). Every example below runs on the small reference samples bundled with the package, so the vignette is network-free.

For the full datasets:

- OTIS → `morie_load_dataset("otisb01")` (downloads via CKAN on first call; subsequent calls hit the local SQLite cache)
- TPS → `morie_fetch_tps("Assault")` (ArcGIS REST)
- SIU → `morie_fetch_siu()` (on-demand scrape of public reports)

See `vignette("mrm-dataset-fetchers")` for the dataset side.

```
library(morie)
b01 <- morie_sample("otis_b01")
b09 <- morie_sample("otis_b09")
tps <- morie_sample("tps_assault")
```

OTIS suite

Placement-count concentration on b09

The b09 long-format file publishes per (fiscal year × placement-count band × gender) counts of individuals in segregation. The callable expands the banded counts using midpoints and returns Hill-MLE Pareto exponent, Gini coefficient, mean placements per individual, and the top-k% concentration share.

```
mrm_otis_placement_concentration(b09)
#>      year n_individuals n_placements mean_per_individual   gini hill_alpha
#> 1  2023         12647         55421         4.382146 0.5331    2.0174
#> 2  2024         10881         47123         4.330760 0.5862    2.1599
#> 3  2025          9608         46893         4.880620 0.6057    2.0880
#> 4 pooled        33136        149437         4.509808 0.5748    2.0814
#> top_pct_share
#> 1         0.2932
#> 2         0.3351
#> 3         0.3215
#> 4         0.3180
```

The values are computed *within fiscal year*: OTIS `UniqueIndividual_ID` has format `YYYY-XXXXX-SG` and is randomly reassigned every fiscal year, so cross-year tracking is invalid by design.

Segregation-duration KM on b01

NumberConsecutiveDays_Segregation is the duration in days of each placement (no censoring — all durations are observed). The callable reports the per-stratum mean, median, q25, and the fraction above the UN Mandela 15-day cutoff.

```
mrm_otis_seg_duration_km(b01)
#>   stratum    n mean_days median_days q25_days pct_above_mandela
#> 1  pooled 1000    3.03         2         3          0.7
#>   median_among_above_mandela
#> 1          54
mrm_otis_seg_duration_km(b01, group_cols = "MentalHealth_Alert")
#>   stratum    n mean_days median_days q25_days pct_above_mandela
#> 1    No 499    2.76         2         3          0.6
#> 2    Yes 501    3.29         2         4          0.8
#>   median_among_above_mandela
#> 1          54.0
#> 2          41.5
```

This callable replaces the misreading of YYYY-XXXX-SG as a persistent person identifier, which produces a spurious cross-year “time-to-readmission” artifact.

Mortification co-occurrence (alert columns)

The three b01 alert flags (MentalHealth_Alert, SuicideRisk_Alert, SuicideWatch_Alert) co-occur to a degree well above independence. The substantive figure is MentalHealth × SuicideRisk Cramer’s V.

```
mrm_otis_mortification_cooccurrence(b01)
#>           alert_a           alert_b    n  chi2 df  p_value crammers_v
#> 1 MentalHealth_Alert SuicideRisk_Alert 1000 33.25 1 8.12e-09 0.1823
#> 2 MentalHealth_Alert SuicideWatch_Alert 1000 12.09 1 5.08e-04 0.1099
#> 3 SuicideRisk_Alert  SuicideWatch_Alert 1000 470.37 1 2.66e-104 0.6858
```

Region locality

Ontario provincial seg/RC placement is overwhelmingly locality-preserving — over 95% of placements remain within the same region in the full b01.

```
# (Region columns are present only in the full b01, not the bundled
# sample; uncomment after morie_load_dataset("otisb01") or
# morie_fetch_tps(...) if needed.)
res <- mrm_otis_region_locality(b01)
print(res$table)
cat("diagonal share:", res$diagonal_share, " V:", res$crammers_v, "\n")
```

Mandela classification

mrm_classify_mandela() shipped in v0.1.14 and remains the canonical Mandela classifier in v0.2.0. It supports three operationalisations:

```

mrm_classify_mandela(b01, denominator = "row") # per-placement
#>   year denominator n_mandela      rate pct n_broader_rc rate_broader
#> 1  2023           362          0 0.000000000 0.00          0 0.000000000
#> 2  2024           337          3 0.008902077 0.89          3 0.008902077
#> 3  2025           301          4 0.013289037 1.33          4 0.013289037
#> 4 pooled          1000          7 0.007000000 0.70          7 0.007000000
mrm_classify_mandela(b01, denominator = "individual_any") # per-person
#>   year denominator n_mandela      rate pct n_broader_rc rate_broader
#> 1  2023           354          0 0.000000000 0.00          0 0.000000000
#> 2  2024           329          3 0.009118541 0.91          3 0.009118541
#> 3  2025           289          4 0.013840830 1.38          4 0.013840830
#> 4 pooled           972          7 0.007201646 0.72          7 0.007201646
mrm_classify_mandela(b01, denominator = "individual_cumulative")
#>   year denominator n_mandela      rate pct n_broader_rc rate_broader
#> 1  2023           354          0 0.000000000 0.00          0 0.000000000
#> 2  2024           329          3 0.009118541 0.91          3 0.009118541
#> 3  2025           289          5 0.017301038 1.73          5 0.017301038
#> 4 pooled           972          8 0.008230453 0.82          8 0.008230453

```

The provincial-canonical 12.5/16.5/20.6 % torture rates from c11 require the c11 aggregate (loaded via `morie_sample("otis_c11")`); see the MRM empirical paper §6.

TPS suite

Levy-flight Hill exponent on inter-event step lengths

Treats consecutive events in chronological order as a single stream and computes the haversine inter-event step length (km). Returns the Hill-MLE exponent restricted to steps above `min_step_km`.

```

mrm_tps_levy_scaling(tps)
#> $n_events
#> [1] 1000
#>
#> $n_steps_tail
#> [1] 995
#>
#> $min_step_km
#> [1] 0.5
#>
#> $hill_alpha
#> [1] 1.3043

```

Moran's I + DBSCAN clustering

Grids the WGS84 extent into a coarse raster, counts events per cell, and computes the global Moran's I via a rook-contiguity matrix. Also runs DBSCAN on the raw lat/long points (rescaled to km) for cluster counts.

```

mrm_tps_moran_clustering(tps, grid_resolution = 20L)
#> $morans_I
#> [1] -0.000138

```

```

#>
#> $morans_z
#> [1] 0.67
#>
#> $dbscan_n_clusters
#> [1] 21
#>
#> $dbscan_n_noise
#> [1] 730
#>
#> $dbscan_largest
#> [1] 82

```

For the high-precision computation on the full 254,378-event Assault file, use the `morie` Python `tps_spatial_advanced` pipeline; the R version is for quick interactive auditing.

Neighbourhood inter-event recurrence

For each HOOD_158 neighbourhood, sorts events chronologically and computes the gap (in days) between consecutive events.

```

head(mrm_tps_neighbourhood_recurrence_km(tps))
#>   hood n_events n_gaps mean_gap_days median_gap_days p25_gap_days p75_gap_days
#> 1  001      17     16    267.06      213.5      101.00      356.75
#> 2  002      12     11    447.91      360.0       74.00      606.00
#> 3  003       5      4    887.75      869.0     557.75     1199.00
#> 4  004       3      2   1781.00     1781.0    1065.50     2496.50
#> 5  005       4      3    776.00      676.0     537.00      965.00
#> 6  006       5      4    845.00      731.0     289.50     1286.50

```

Hawkes manifest loader

`mrm_tps_load_hawkes_refit(path)` reads `paper_hawkes_refit.json` (the per-category Hawkes refit table from the MRM empirical paper §7.1-7.2) and returns it as a tidy data.frame. Skipped here because the JSON is in the MOIRAIS data manifest, not bundled with the package.

SIU suite

The SIU callables operate on the SIU.csv file produced by `morie_fetch_siu()` (an on-demand scraper of the public Director's Reports). The scraped corpus is not shipped, but the callables themselves do not depend on shipped data.

```

siu_path <- morie_fetch_siu()
siu <- read.csv(siu_path)
res <- mrm_siu_case_to_decision_km(siu)
print(res$pooled)
head(res$by_service[order(-res$by_service$n),])
mrm_siu_per_service_rate(siu)
mrm_siu_outcome_classifier(siu)

```

The verified pooled median in our test snapshot is **120 days from incident to Director’s decision** (n = 1,711 cases). Per-service medians cluster tightly around 120, indicating a system-wide processing cadence rather than a per-jurisdiction effect.

References

- MRM theoretical paper — Ruhela (2026), *The McNamara-Ruhela-Medina framework*, Zenodo 10.5281/zenodo.20096075.
- MRM empirical paper — Ruhela (2026, in preparation), [papers/morie-empirical-paper/](#).
- OTIS data dictionary — [data.ontario.ca/dataset/data-on-inmates-in-ontario](#).
- Toronto Police Open Data — [data.torontopolice.on.ca/](#).
- SIU public Director’s Reports — [siu.on.ca/en/case_directors_reports.php](#).